

PS 2 Biomechanical Fingerprinting: From Tissue Modelling to Synthetic Gel Design ‹ Advanced ›

Designed and mentored by **Dr. Kailasham Ramalingam**, Assistant Professor, Department of Chemical Engineering, IIT Indore.

Background. The development of synthetic organs and surgical simulators requires highly accurate descriptions of human tissue behaviour. Tissues like the myocardium or arterial walls are hyperelastic, anisotropic, and non-linear. By determining a tissue's material fingerprint — the unique set of constants defining its stress-stretch relationship — we can engineer synthetic hydrogels that mimic these exact mechanical properties.

The Objective. Teams will compile experimental tissue data, use an existing automated Python pipeline to extract the material fingerprint, and subsequently use LAMMPS to design a 3D polymer gel architecture that mimics this exact force-extension behaviour.

Technical Tasks.

- **Literature Mining:** Extract stress-stretch (σ - λ) data for a distinct human tissue — cardiac tissue, skin, or arterial wall — from peer-reviewed biomechanics journals.
- **Material Fingerprinting:** Do not reinvent the wheel. Utilise the existing open-source material fingerprinting Python pipeline developed by Moritz Flaschel and co-workers to fit the experimental data to appropriate hyperelastic models (Demiray, Mooney–Rivlin, or Holzapfel–Gasser–Ogden) and extract the material constants.
- **In-Silico Gel Assembly:** Construct a 3D coarse-grained, crosslinked polymer network representing a synthetic gel within LAMMPS.
- **Biomechanical Mimicry:** Perform simulated tensile testing on the LAMMPS gel using fix deform. Iteratively adjust the gel's architectural parameters — crosslink density, chain length between crosslinks — until the simulated force-extension curve matches the fingerprinted tissue data.

Resources.

- Flaschel, Kumar & De Lorenzis — hyperelastic fingerprinting pipeline paper and code, CMAME 2021
- *Nonlinear Solid Mechanics* — Holzapfel
- YouTube: *Continuum Mechanics* — Prof. Bob McGinty
- *Computer Simulation of Liquids* — Allen & Tildesley
- LAMMPS documentation: fix deform, NPT ensemble, virial stress tensor

Deliverables.

- Output logs and validation plots from the Flaschel fingerprinting pipeline.
- LAMMPS data files and input scripts for gel construction and tensile testing.
- A comparative plot overlaying the biological tissue's experimental data with the LAMMPS gel's simulated mechanical response.
- GitHub repository and technical report.

Team Size. 4–5 members.